

A REPORT
ON
**TRANSFER MATRIX - PSR SCOUTING
INTELLIGENCE**

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Abstract

Modern football is increasingly dominated by clubs with massive financial resources, making it difficult for mid-tier clubs to compete while adhering to strict Profitability and Sustainability Rules (PSR). This project, titled *Transfer Matrix*, addresses this financial disparity by proposing an AI-powered scouting system. Utilizing Principal Component Analysis (PCA) and Cosine Similarity, the system identifies highly correlated, cost-effective "statistical twins" of elite football players. Furthermore, it integrates Generative AI models (Gemini and Ollama) to produce automated, natural-language scouting reports. This report details the system architecture, methodology, performance metrics, and practical applications of the *Transfer Matrix* platform.

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Chapter 1

Project Overview & Objective

1.1 Project Title

Transfer Matrix - PSR Scouting Intelligence

1.2 Problem Statement

In the contemporary landscape of professional football, there exists a massive financial disparity. Elite clubs possess the capital to acquire hundred-million-dollar players, whereas smaller and mid-tier clubs struggle to remain competitive. Furthermore, all clubs are bound by strict Profitability and Sustainability Rules (PSR), which severely limit financial losses over a rolling period. The exact problem this system solves is the inability of smaller clubs to scout and acquire elite-level talent without breaching their financial constraints.

1.3 Objective

The objective is to build an AI-powered web platform that automates and enhances the efficiency of football scouting using machine learning algorithms. By mathematically identifying "statistical twins"—players who produce identical underlying numbers on the pitch but cost a fraction of the price—the system allows clubs to replace elite talent on a shoestring budget, thereby maintaining team performance while remaining financially compliant.

Chapter 2

System Architecture & Tech Stack

2.1 Dataset

The system utilizes a comprehensive dataset comprising statistical data for 230 players across the "Big 5" European leagues. The data was aggregated from reputable football data providers and underwent rigorous preprocessing. Null values were handled, and all performance metrics were normalized per 90 minutes to ensure a fair comparison across players with varying amounts of playing time.

2.2 Algorithms/Models

The core matching engine relies on two primary mathematical frameworks:

- **Principal Component Analysis (PCA):** Used for dimensionality reduction. It compresses hundreds of complex player metrics into principal components that capture the most significant variance in playing styles.
- **Cosine Similarity:** Applied on the PCA-reduced feature space to calculate the distance and similarity score between players, effectively finding the closest statistical matches.
- **Generative AI (LLMs):** Google's Gemini API and local Ollama (Llama 3) models are employed to analyze the statistical outputs and generate natural-language scout reviews.

2.3 UI & Deployment

The user interface was developed using Vanilla HTML, CSS, and JavaScript, prioritizing a dependency-free, lightning-fast experience. It features a premium, glassmorphism aesthetic with dynamic micro-animations. The backend API is powered by Python and FastAPI, handling the heavy computational loads and model inference before serving the results to the frontend.

Chapter 3

Methodology & Workflow

3.1 Data Collection

Raw statistical data, including passing, shooting, defensive actions, and possession metrics, was collected for players in the top 5 European leagues. Market values and contract data were also sourced to enable PSR budget filtering.

3.2 Data Preprocessing

The data preprocessing pipeline involved several critical steps:

- **Cleaning:** Removing duplicate entries and handling missing or null values using median imputation techniques.
- **Normalization:** Scaling features using standard scalers so that metrics with different ranges (e.g., pass completion percentage vs. total distance covered) do not disproportionately bias the similarity calculations.

3.3 Model Training

The training and execution phase involved:

- **Dimensionality Reduction:** Applying PCA to the normalized dataset to extract the top components that explain the vast majority of the variance.

- **Hyperparameter Tuning:** Adjusting the number of principal components to balance computational efficiency with the preservation of statistical nuance.

3.4 Testing & Validation

Validation was conducted by evaluating precision and recall metrics based on known player archetypes. Similarity scores were cross-referenced with domain expert evaluations to ensure the mathematical twins accurately reflected similar on-pitch roles and output.

Chapter 4

Results & Performance Metrics

4.1 Accuracy/Evaluation

The system successfully identifies highly correlated statistical matches in near real-time. The model achieved a robust output, consistently predicting "statistical twins" with similarity scores exceeding 90% for high-profile target players.

4.2 Output Interface

The user interacts with a highly dynamic web dashboard. The workflow is as follows:

1. The user selects a target player and sets a PSR budget ceiling using a range slider.
2. The system computes the similarities and returns a shortlist of affordable "twins."

Target Player	Recommended Twin	Similarity Score	Cost Savings
Declan Rice	Yousouf Fofana	94.2%	£70M
Rodri	Martin Zubimendi	91.8%	£45M
Jude Bellingham	Teun Koopmeiners	89.5%	£60M

Table 4.2: Sample Twin Recommendations and Financial Impact

Table 4.2 demonstrates the potential financial savings of adopting the system’s recommendations while maintaining a similarity score above 89%.

Chapter 5

Practical Applications & Future Scope

5.1 Real-World Impact

The primary application of this project lies in professional football scouting and recruitment. It can be directly utilized by data analysts, chief scouts, and sporting directors at professional clubs to optimize their transfer budgets. By identifying undervalued talent, clubs can ensure compliance with PSR regulations while maintaining a highly competitive squad.

5.2 Future Enhancements

While the current system is highly capable, several future enhancements can scale its utility:

- **Integration of Live Match Data:** Incorporating real-time tracking and event data to refine the statistical models further.
- **Global League Expansion:** Scaling the dataset beyond the Big 5 leagues to uncover hidden gems in lower-tier or obscure global leagues.
- **Commercial Valuation Integration:** Combining on-pitch statistical scouting with off-pitch commercial valuation models (e.g., social media reach) to calculate a player's True Return on Investment (ROI).

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Appendix A

PCA Mathematical Formulation

The Principal Component Analysis involves finding the eigenvalues λ and eigenvectors ν of the data's covariance matrix Σ .

$$\Sigma\nu = \lambda\nu \tag{A.1}$$

The eigenvectors corresponding to the largest eigenvalues represent the principal components that capture the maximum variance in player statistics.

Appendix B

Cosine Similarity Formulation

This appendix outlines the Cosine Similarity metric utilized to determine the statistical proximity between two players. Given two vectors A and B representing the principal components of two players, the cosine similarity is calculated as the dot product of the vectors divided by the product of their magnitudes:

$$\text{Similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (\text{B.1})$$

A value closer to 1 indicates a highly correlated "statistical twin," while a value closer to 0 indicates orthogonally distinct playing styles.